

HKCEE 1992

Mathematics II

Label 1 $\frac{1}{a} + \frac{1}{b} =$

- A. $\frac{a+b}{ab}$
 B. $\frac{ab}{a+b}$
 C. $\frac{1}{ab}$
 D. $\frac{2}{a+b}$
 E. $\frac{1}{a+b}$

Figure 1 Density 0

Label 2 If $a = 1 - \frac{1}{1-b}$, then $b =$

- A. $1 - \frac{1}{1-a}$
 B. $1 - \frac{1}{1+a}$
 C. $1 + \frac{1}{1-a}$
 D. $1 + \frac{1}{1+a}$
 E. $-1 + \frac{1}{1+a}$

Label 3 For what value(s) of x does the equality $\frac{(x+1)(x-2)}{x-2} = x+1$ hold?

- A. -1 only
 B. 2 only
 C. Any value
 D. Any value except -1
 E. Any value except 2

Label 4 $\frac{\sqrt{5}+1}{\sqrt{5}-1} - \frac{\sqrt{5}-1}{\sqrt{5}+1} =$

- A. 0
 B. $\frac{1}{2}$
 C. 3
 D. $\sqrt{5}$
 E. $\frac{1}{2} + \sqrt{5}$

Label 5 If $\log_{10} b = 1 + \frac{1}{2} \log_{10} a$, then $b =$

- A. $10\sqrt{a}$
 B. $10 + \sqrt{a}$
 C. $5a$
 D. $\frac{a}{2}$
 E. $1 + \frac{a}{2}$

Label 6 Which of the following is a factor of $4(a+b)^2 - 9(a-b)^2$?

- A. $5b-a$
 B. $5b+a$
 C. $a=b$
 D. $3b-5a$
 E. $3a-5b$

Label 7 If $\frac{a}{b} = \frac{c}{d} = k$ and a, b, c, d are positive, then which of the following must be true?

- A. $\frac{a+c}{b+d} = k$
 B. $ab = cd = k$
 C. $ac = bd = k$
 D. $a = c = k$
 E. $\frac{ac}{bd} = k$

Label 8

Simplify $\frac{\overbrace{n \times n \times \cdots \times n}^{n \text{ times}}}{\underbrace{n + n + \cdots + n}_{n \text{ terms}}}$

- A. n^{n-2}
- B. $\frac{n}{n^2}$
- C. $n = 2$
- D. $\frac{n}{2}$
- E. 1

Label 9 If a and b are greater than 1, which of the following statements is/are true?

- I. $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$
- II. $(a^{-1} + b^{-1})^{-1} = a + b$
- III. $a^2 b^3 = (ab)^6$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. None of them

Label 10 If $a : b = 2 : 3$, $a : c = 3 : 4$ and $b : d = 5 : 2$, find $c : d$.

- A. 1 : 5
- B. 16 : 45
- C. 10 : 3
- D. 20 : 9
- E. 5 : 1

Label 11 Suppose x varies directly as y^2 and inversely as z . Find the percentage increase of x when y is increased by 20% and z is decreased by 20%.

- A. 15.2%
- B. 20%
- C. 50%
- D. 72.8%
- E. 80%

Label 12

A sum of \$ 10 000 is deposited at 4% p.a., compounded yearly. Find the interest earned *in the second year*.

- A. \$16
- B. \$400
- C. \$416
- D. \$800
- E. \$816

Label 13

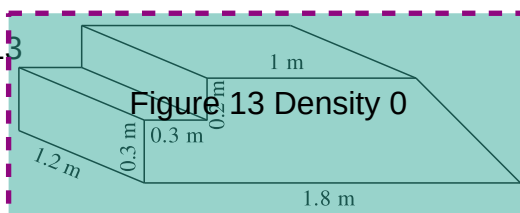


Figure 13 Density 0

The figure shows a solid platform with steps on one side and a slope on the other. Find its volume.

- A. 0.75 m^3
- B. 0.84 m^3
- C. 0.858 m^3
- D. 1.008 m^3
- E. 1.608 m^3

Label 14

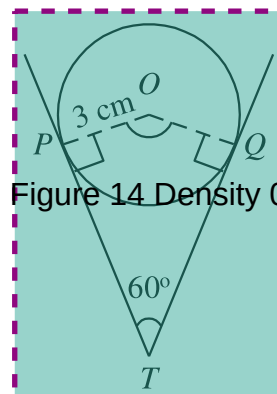


Figure 14 Density 0

In the figure, TP and TQ are tangent to the circle of radius 3 cm. Find the length of the minor arc PQ.

- A. $3\pi \text{ cm}$
- B. $2\pi \text{ cm}$
- C. $\frac{3\pi}{2} \text{ cm}$
- D. $\pi \text{ cm}$
- E. $\frac{\pi}{2} \text{ cm}$

Label 15

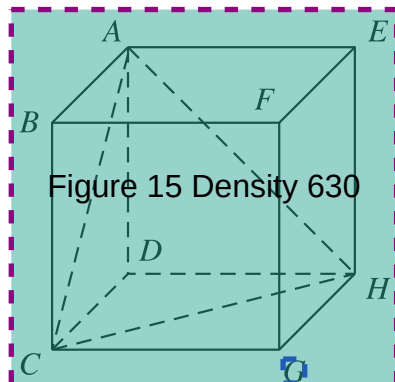


Figure 15 Density 630

Find the ratio of the volume of the tetrahedron $ACHD$ to the volume of the cube $ABCDEFGH$ in the figure.

- A. $1 : 8$
- B. $1 : 6$
- C. $1 : 4$
- D. $1 : 3$
- E. $1 : 2$

Label 16

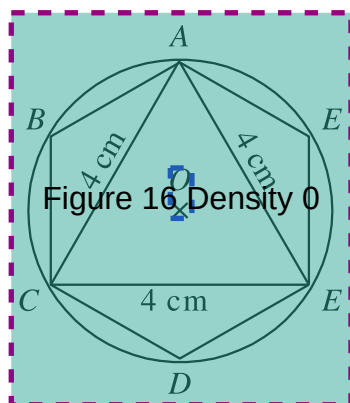


Figure 16 Density 0

In the figure, the equilateral triangle ACE of side 4 cm is inscribed in the circle. Find the area of the inscribed regular hexagon $ABCDEF$.

- A. $8\sqrt{3} \text{ cm}^2$
- B. $8\sqrt{2} \text{ cm}^2$
- C. $4\sqrt{3} \text{ cm}^2$
- D. $4\sqrt{2} \text{ cm}^2$
- E. 16 cm^2

Label 17

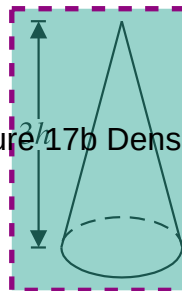


Figure 17b Density 84

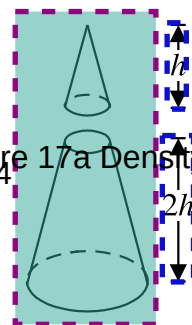


Figure 17a Density 0

In the figure, a cone of height $3h$ is cut by a plane parallel to its base into a smaller cone of height h and a frustum. Find the ratio of the volume of the smaller cone to the volume of the frustum.

- A. $1 : 27$
- B. $1 : 26$
- C. $1 : 9$
- D. $1 : 8$
- E. $1 : 7$

Label 18 The greatest value of $1 - 2\sin \theta$ is

- A. 5
- B. 3
- C. 1
- D. 0
- E. -1

Label 19

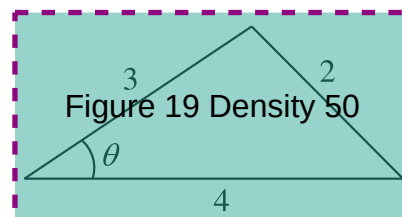


Figure 19 Density 50

In the figure, find $\cos \theta$.

- A. $\frac{1}{4}$
- B. $\frac{11}{16}$
- C. $\frac{3}{4}$
- D. $\frac{7}{8}$

E. $\frac{\sqrt{77}}{9}$

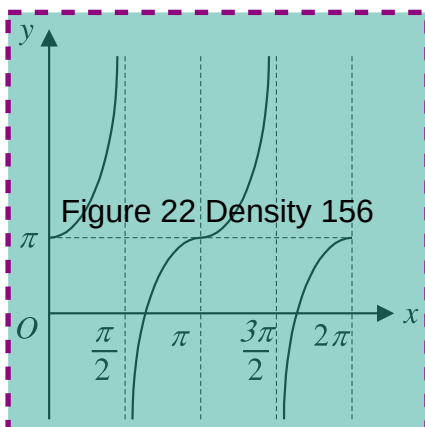
Label 20 In which two quadrants will the solution(s) of $\sin \theta \cos \theta < 0$ lie?

- A. In quadrants I and II only
- B. In quadrants I and III only
- C. In quadrants II and III only
- D. In quadrants II and IV only
- E. In quadrants III and IV only

Label 21 If $A + B + C = 180^\circ$, then $1 + \cos A \cos (B + C) =$

- A. 0
- B. $\sin^2 A$
- C. $1 + \cos^2 A$
- D. $1 + \sin A \cos A$
- E. $1 - \sin A \cos A$

Label 22



The figure shows the graph of the function

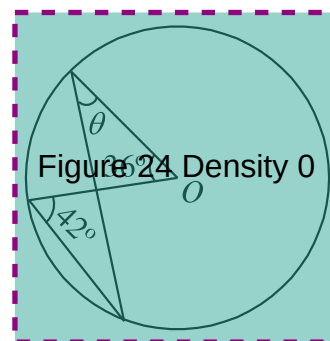
- A. $\tan(x + \pi)$
- B. $\tan(x - \pi)$
- C. $\pi \tan x$
- D. $\pi + \tan x$
- E. $\pi - \tan x$

Label 23 Which of the following equations has/have solutions?

- I. $2 \cos^2 \theta - \sin^2 \theta = 1$
- II. $2 \cos^2 \theta - \sin^2 \theta = 2$
- III. $2 \cos^2 \theta - \sin^2 \theta = 1$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

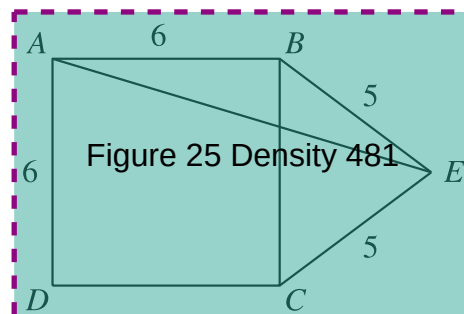
Label 24



In the figure, O is the centre of the circle. Find θ

- A. 42°
- B. 36°
- C. 24°
- D. 21°
- E. 18°

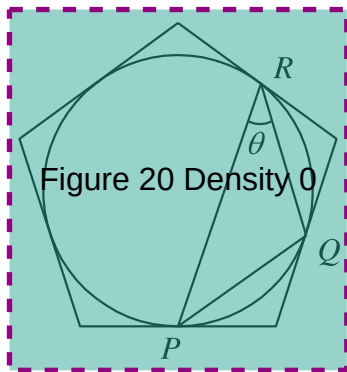
Label 25



In the figure, ABCD is a square with side 6. If $BE = CE = 5$, find AE.

- A. $\sqrt{61}$
- B. 9
- C. 10
- D. $6\sqrt{3}$
- E. $\sqrt{109}$

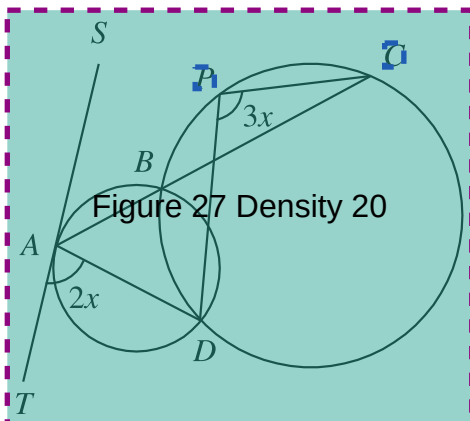
Label 20
26



In the figure, the circle is inscribed in a regular pentagon. P , Q and R are points of contact. Find θ .

- A. 30°
- B. 32°
- C. 35°
- D. 36°
- E. 45°

Label 27
27



In the figure, ST is a tangent to the smaller circle. ABC is a straight line. If $\angle TAD = 2x$ and $\angle DPC = 3x$, find x .

- A. 30°
- B. 36°
- C. 40°
- D. 42°
- E. 45°

Label 28
28

If the two lines $2x - y + 1 = 0$ and $ax + 3y - 1 = 0$ do not intersect, then a is

- A. -6
- B. -2

- C. 2
- D. 3
- E. 6

Label 29
29

If $0 < k < h$, which of the following circles intersect(s) the y -axis?

- I. $(x - h)^2 + (y - k)^2 = k^2$
- II. $(x - h)^2 + (y - k)^2 = h^2$
- III. $(x - h)^2 + (y - k)^2 = h^2 + k^2$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

Label 30
30

If the line $y = mx + 3$ divides the circle $x^2 + y^2 - 4x - 2y - 5 = 0$ into two equal parts, find m .

- A. $\frac{1}{4}$
- B. $-\frac{1}{4}$
- C. 0
- D. $\frac{5}{4}$
- E. 2

Label 31
31

The mid-points of the sides of a triangle are $(3, 4)$, $(2, 0)$ and $(4, 2)$. Which of the following points is a vertex of the triangle?

- A. $(3.5, 3)$
- B. $(3, 2)$
- C. $(3, 1)$
- D. $(1.5, 2)$
- E. $(1, 2)$

Label 32
32

The table shows the mean marks of two classes of students in a

	Number of students	Mean mark
Class A	36	72
Class B	42	54

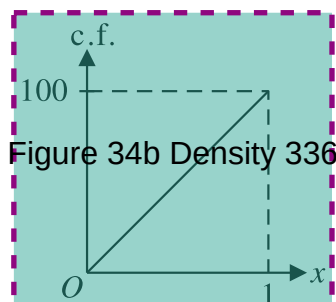
A student in Class A has scored 91 marks. It is found that his score was wrongly recorded as 19 in the calculation of the mean mark for Class A in the above table. Find the correct mean mark of the 80 students in the two classes.

- A. 61.65
- B. 62.55
- C. 63
- D. 63.45
- E. 63.9

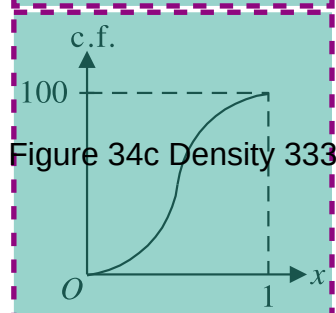
Label 33 Two cards are drawn randomly from five cards A, B, C, D and E. Find the probability that card A is drawn while card C is not.

- A. $\frac{3}{25}$
- B. $\frac{3}{20}$
- C. $\frac{4}{25}$
- D. $\frac{6}{25}$
- E. $\frac{3}{10}$

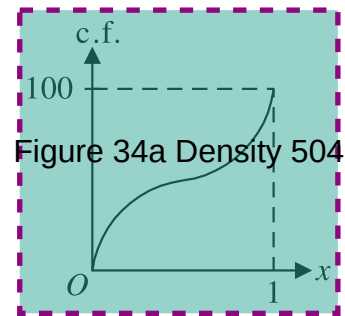
Label 34



II.



III.



The figure shows the cumulative frequency curves of three distributions. Arrange the three distributions in the order of their standard deviations, from the smallest to the largest.

- A. I, II, III
- B. I, III, II
- C. II, I, III
- D. II, III, I
- E. III, I, II

Label 35 If the quadratic equation $ax^2 - 2bx + c = 0$ has two equal roots, which of the following is/are true?

- I. a, b, c form an arithmetic progression.
- II. a, b, c form a geometric progression.
- III. Both roots are $\frac{b}{a}$.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

Label 36 Which of the following intervals must contain a root of $2x^3 = x^2 - x + 3 = 0$?

- I. $-1 < x < 0$
- II. $0 < x < 2$
- III. $1 < x < 3$

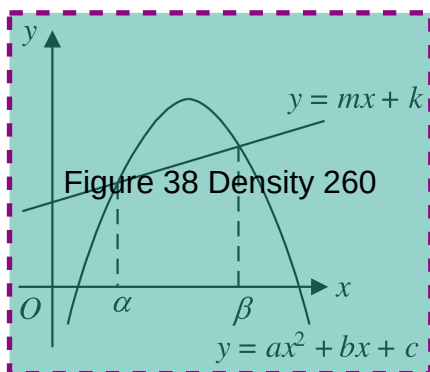
- A. I only
- B. II only
- C. III only

- D. I and II only
E. II and III only

Label 37 How many integers x satisfy the inequality $6x^2 - 7x - 20 \leq 0$?

- A. 0
B. 1
C. 2
D. 3
E. 4

Label 38



From the figure, if $\alpha \leq x \leq \beta$, then

- A. $ax^2 + (b - m)x + (c - k) \leq 0$
B. $ax^2 + (b - m)x + (c - k) < 0$
C. $ax^2 + (b - m)x + (c - k) = 0$
D. $ax^2 + (b - m)x + (c - k) > 0$
E. $ax^2 + (b - m)x + (c - k) \geq 0$

Label 39 Under which of the following conditions *must* the mean of n consecutive positive integers also be an integer?

- A. n is any positive integer
B. n is any positive odd integer
C. n is any positive even integer
D. n is any multiple of 3
E. n is the square of any positive integer

Label 40 The L.C.M. of P and Q is $12ab^3c^2$. The L.C.M. of X , Y and Z is $30a^2b^3c$. What is the L.C.M. of P , Q , X , Y and Z ?

- A. $360a^3b^6c^3$
B. $60a^2b^3c^2$

- C. $60ab^3c^3$
D. $6a^2b^3c$
E. $6ab^3c$

Label 41 If a polynomial $f(x)$ is divisible by $x - 1$, then $f(x - 1)$ is divisible by

- A. $x - 2$
B. $x + 2$
C. $x - 1$
D. $x + 2$
E. x

Label 42 Find the $(2n)$ th term of G.P. $-\frac{1}{2}, 1, -2, 4, \dots$

- A. 2^{2n}
B. -2^{2n}
C. -2^{2n-3}
D. 2^{2n-2}
E. -2^{2n-2}

Label 43 If the price of an orange rises by \$1, then 5 fewer oranges could be bought for \$100. Which of the following equations gives the original price \$ x of an orange?

- A. $\frac{100}{x+1} = 5$
B. $\frac{100}{x+1} - \frac{100}{x} = 5$
C. $\frac{100}{x} - \frac{100}{x+1} = 5$
D. $\frac{100}{x-1} - \frac{100}{x} = 5$
E. $\frac{100}{x} - \frac{100}{x-1} = 5$

Label 44 By selling an article at 10% discount off the marked price, a shop still makes 20% profit. If the cost price of the article is \$19 800, then the marked price is

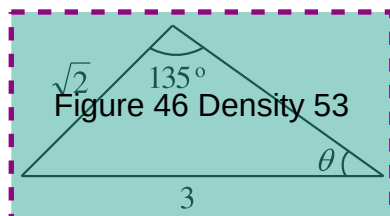
- A. \$21 600

- B. \$26 136
- C. \$26 400
- D. \$27 225
- E. \$27 500

Label 45
45 Coffee A and coffee B are mixed in the ratio $x : y$ by weight. A costs \$50/kg and B costs \$40/kg. If the cost of A is increased by 10% which that of B is decreased by 15%, the cost of the mixture per kg remains unchanged. Find $x : y$.

- A. 2 : 3
- B. 5 : 6
- C. 6 : 5
- D. 3 : 2
- E. 55 : 34

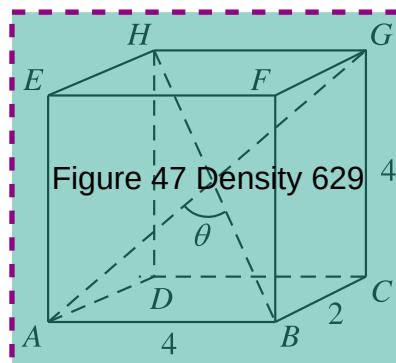
Label 46
46



In the figure, find $\tan \theta$

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{8}}$
- C. $\frac{3}{8}$
- D. $\sqrt{\frac{2}{7}}$
- E. $\frac{1}{\sqrt{2}}$

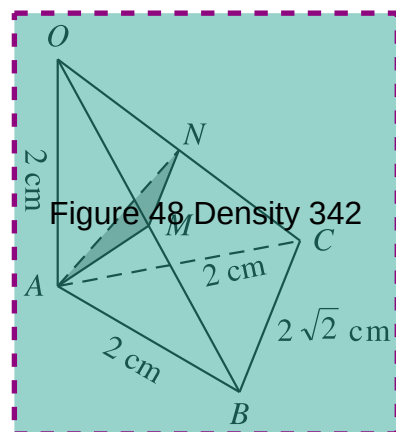
Label 47
47



In the figure, if θ is the angle between the diagonals AG and BH of the cuboid, then

- A. $\sin \frac{\theta}{2} = \frac{2}{3}$
- B. $\sin \frac{\theta}{2} = \frac{3}{4}$
- C. $\sin \frac{\theta}{2} = \frac{1}{3}$
- D. $\sin \theta = \frac{2}{3}$
- E. $\sin \theta = \frac{3}{4}$

Label 48
48



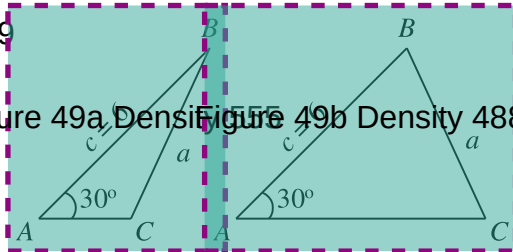
In the figure, OA is perpendicular to the plane ABC. $OA = AB = AC = 2$ cm and $BC = 2\sqrt{2}$ cm. If M and N are the mid-point of OB and OC respectively, find the area of $\triangle AMN$.

- A. $\frac{1}{2}$ cm²
- B. 1 cm²

- C. $\sqrt{2} \text{ cm}^2$
 D. $\frac{\sqrt{3}}{2} \text{ cm}^2$
 E. $\sqrt{3} \text{ cm}^2$

Label 49
49

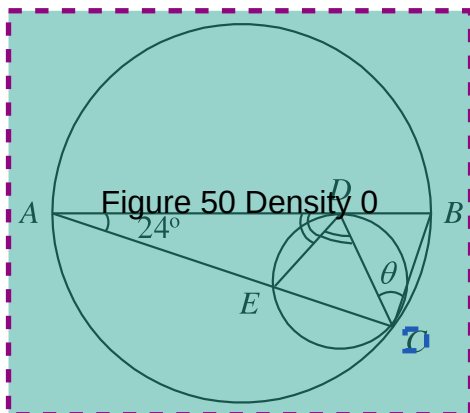
Figure 49a Density 498
Figure 49b Density 488



In $\triangle ABC$, $\angle A = 30^\circ$, $c = 6$. If it is possible to draw two distinct triangles as shown in the figure, find the range of values of a .

- A. $0 < a < 3$
 B. $0 < a < 6$
 C. $3 < a < 6$
 D. $a > 3$
 E. $a > 6$

Label 50
50



In the figure, the two circles touch each other at C . The diameter AB of the bigger circle is tangent to the smaller circle at D . If DE bisects $\angle ADC$, find θ .

- A. 24°
 B. 38°
 C. 45°
 D. 52°
 E. 66°

Label 51
51

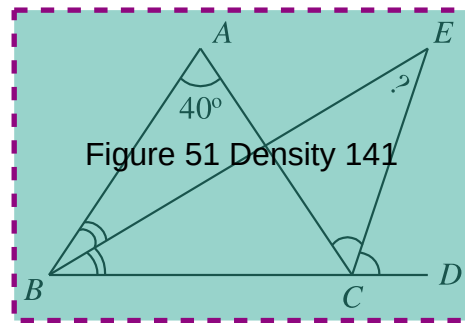


Figure 51 Density 141

In the figure, EB and EC are the angle bisectors of $\angle ABC$ and $\angle ACD$ respectively. If $\angle A = 40^\circ$, find $\angle E$.

- A. 20°
 B. 25°
 C. 30°
 D. 35°
 E. 40°

Label 52
52

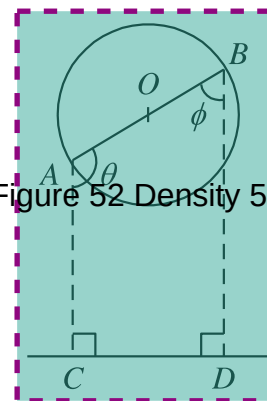


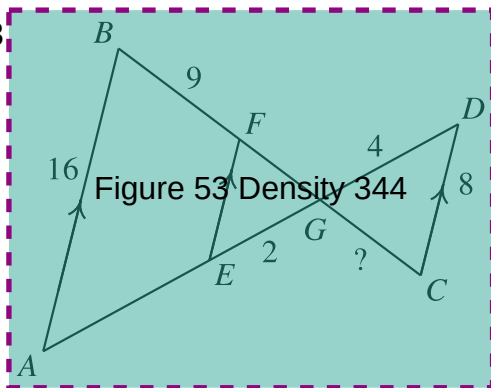
Figure 52 Density 521

In the figure, O is the centre of the circle. If the diameter AOB rotates about O , which of the following is/are constant?

- I. $\theta + \phi$
 II. $AC + BD$
 III. $AC \times BD$

- A. I only
 B. II only
 C. III only
 D. I and II only
 E. II and III only

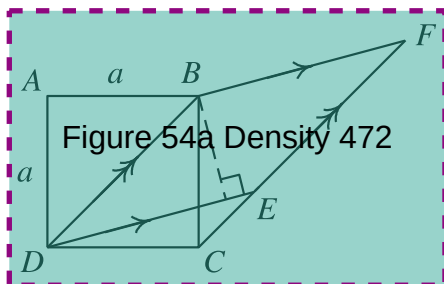
Label 53



In the figure, $AB = 16$, $CD = 8$, $BF = 9$, $GD = 4$, $EG = 2$. Find GC .

- A. 4.5
- B. 5
- C. 6
- D. 8
- E. 10

Label 54



In the figure, $ABCD$ is a square of side a and $BDEF$ is a rhombus. CEF is a straight line. Find the length of the perpendicular from B to DE .

- A. $\frac{1}{2}a$
- B. $2a$
- C. $\frac{a}{\sqrt{3}}$
- D. $\frac{\sqrt{3}}{2}a$
- E. a

Figure 54b Density 1059